

Product Requirements Document (PRD) & System Architecture

Project: Global Macro-Sentiment Tracker (Agile PostgreSQL) **Version:** 3.0 (Master Final)

1. Executive Summary & Requirements

1.1 Goal

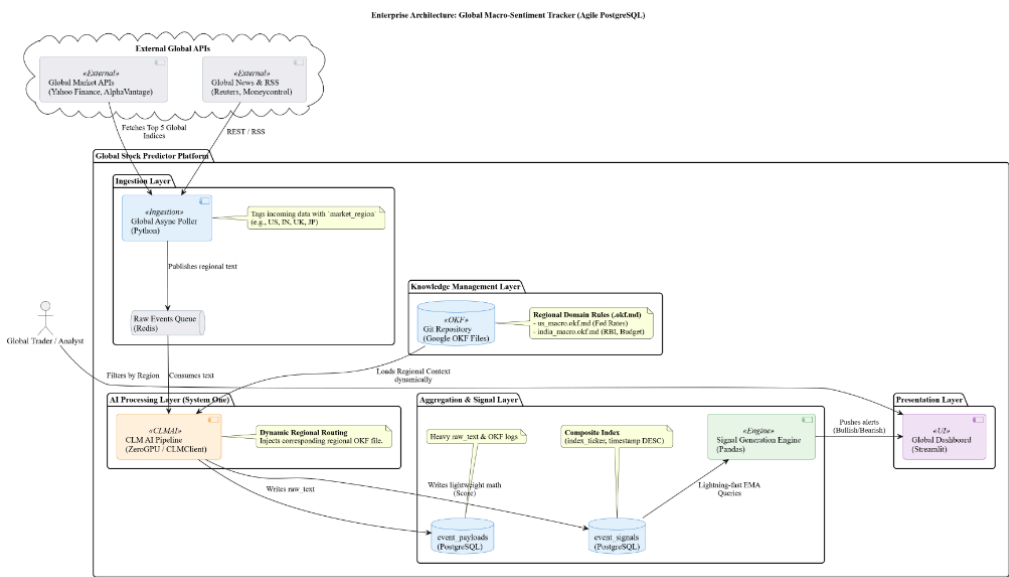
Build an ultra-low-latency, AI-driven stock prediction pipeline that evaluates global financial news in real-time, filters noise, and triggers automated trading signals. The system explicitly tracks the **Top 5 Global Indices** (e.g., S&P 500, Nifty 50, FTSE 100) to predict major market trends with a high signal-to-noise ratio.

1.2 Core Requirements

- **Target Scope:** Global Index Tracking across regions (US, IN, UK, JP).
- **Ingestion:** Reliable async polling from global data providers (Yahoo Finance, Global RSS).
- **Processing:** Contrastive Language Models (ZeroGPU CLM-8B, INT4) for deterministic System-One scoring (Choice, Score, Noul).
- **Knowledge Layer:** Strict decoupling of static, region-specific trading rules using Google OKF.
- **Signaling:** 4-hour Exponential Moving Averages (EMA) to emit directional signals.

2. High Level Design (HLD) & Architecture

Below is the container architecture illustrating the ingestion pipeline, the dynamic regional routing of the Google OKF rules, and the vertically partitioned PostgreSQL database.



3. Database Architecture & Features (Agile PostgreSQL)

To guarantee sub-millisecond query latency for the Signal Engine during high-frequency ingestion, the database utilizes **Vertical Partitioning** and advanced indexing on Supabase (Managed PostgreSQL).

3.1 Vertical Partitioning Strategy

Instead of a single bloated table, data is split into two tables sharing a 1-to-1 relationship via UUIDs:

- **event_signals (The Math Layer):** Extremely lightweight. Stores only the critical mathematical points needed by the Signal Engine (`index_ticker`, `market_region`, `timestamp`, `sentiment_score`).
- **event_payloads (The Document Layer):** Stores the heavy data (`raw_text`, `applied_okf_rules`, JSON metadata). Isolating this prevents bulky text from slowing down the Signal Engine's sequential scans.

3.2 Indexing & Performance Features

- **Composite Indexing:** A composite B-Tree index is applied to (`index_ticker`, `timestamp DESC`) on `event_signals`. This perfectly matches the exact read pattern of the Signal Engine, allowing it to instantly fetch the last 4 hours of sentiment for a specific index.
- **Time-Series Ready:** The `timestamp` column utilizes a BRIN (Block Range Index) which consumes 99% less memory than standard indexes, massively accelerating the ingestion worker's massive bulk inserts.
- **Connection Pooling:** Supabase's native PgBouncer handles the connection pooling, preventing TCP exhaustion when the Python async poller sends hundreds of concurrent inserts.

4. Implementation Plan & SDLC Practices

4.1 Software Layers

1. **Global Ingestion Worker:** Polls REST APIs, tagging payloads with a `market_region`.
2. **Regional AI Engine:** Dynamically injects `us_macro.okf.md` or `india_macro.okf.md` into the AI context based on the region.
3. **Signal Engine:** Fast vector math via Pandas, triggered via Cron jobs.
4. **UI (Streamlit):** Real-time dashboard with regional filtering.

4.2 Recommended SDLC Practices

- **GitOps for Knowledge:** Any regional `.okf.md` updates require a Pull Request review to ensure human oversight on AI logic.
- **CI/CD:** GitHub Actions to run PyTest verification on the OKF parsing engine.

5. Deployment Strategy & Cost Estimates

5.1 Architecture (Serverless / \$0 CapEx)

- **AI Inference:** Hosted on **Hugging Face "ZeroGPU" Spaces** (Dynamic free A100 slices using INT4 quantized models).
- **Dashboard UI:** **Streamlit Community Cloud**.
- **Database:** **Supabase** (Free Tier PostgreSQL).
- **Ingestion Worker:** Containerized via Docker on **Render/Railway**.

5.2 Estimated Monthly Costs

- **Market Data / News:** \$0/mo (Free Tiers)
 - **Database / Hosting:** \$0/mo (Free Tiers)
 - **AI Compute:** \$0/mo (ZeroGPU)
 - **Total Estimated Cost:** **\$0.00 / month**.
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6. Testing & Benchmarking Approach

6.1 Quality Assurance (QA)

- **Integration Testing:** Ensure the CLM Engine correctly maps Indian news to `india_macro.okf.md` and US news to `us_macro.okf.md`.

6.2 Benchmarking Strategy

- **Latency Budget:** The pipeline must process a headline from Ingestion to the `event_payloads` table in **< 150ms**.
- **Win Rate:** Backtest 3 months of Nifty 50 and S&P 500 news. Ensure directional accuracy (Win Rate) exceeds **55%** before transitioning from Alpaca Paper Trading to Live Money.